Stephen Welch

Gainesville, VA

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Education

B.S. in Computer Engineering — *Virginia Polytechnic Institute*

Graduating May 2023 | GPA: 3.29 | Focus in Controls, Robotics, and Autonomy, Minor in Computer Science — *Aug. 2019 - May 2023*

Relevant Courses: AI & Engineering Applications (ECE 4524), Machine Learning (CS 4824), Software Design and Data Structures (CS 3114), Linux Kernel Programming (CS 4424)

Battlefield High School — *Battlefield High School*

Information Technology Program — *Aug. 2015-2019*

GPA: 4.38

Skills

Tools	Git, Gradle, Jetbrains IDEs, Visual Studio, GNUPlot, LaTeX
Frontend	JavaFX, Swing, Qt, ASP.NET
Languages	Python, Java 8, C++, C#, SQL

Work Experience

Machine Learning Engineer (Part-time) — *Alexandria, VA*

Shield AI — *Aug. 2021-Present*

- Key contributor to development of collaborative autonomy for counter-IADS missions
- Utilized multi-agent reinforcement learning, task allocation, and path planning algorithms to produce flexible and robust teaming behaviours in **Python**
- Sim-to-sim transfer research utilizing model-based reinforcement learning in **PyTorch**

AI Intern — *Alexandria, VA*

Shield AI — *Dec. 2020-Aug. Present*

- Designed and developed multi-agent simulation framework in **Python** in collaboration with senior engineers and other interns. Currently in use across multiple government contracts and internal projects
- Trained sim-to-real reinforcement learning agents to track fixed-wing drone flight paths in JSBSim using a custom **PyTorch**-based RL framework
- Trained reinforcement learning agents in **RLLib** to engage in air-to-air combat scenarios as fixed-wing UAVs simulated in JSBSim. Implemented weapon engagement zones and rules of engagement for combat environment
- Created replay visualizer using Tacview and X-Plane in **Lua** to help F-22, F-35, F-16 pilots evaluate air-to-air engagements with autonomous agents

Volunteer Researcher — *Blacksburg, VA*

Terrestrial Robotics Engineering and Controls Lab — *Jan. 2020 - Present*

- Developed
- Developed custom communications and controls code for the PANDORA humanoid robot using the **Java**-based IHMC humanoid robotics stack
- Implement force-torque mapping to enable torque control of linearly-actuated humanoid in **Python** and **Java**

Enterprise Applications Intern — *Herndon, VA*

Serco North America — *Mar. 2020 - Aug. 2020*

- Wrote utilities to verify website integrity using **Java**, **JavaFX**, **Swing**, and **Selenium** with interactive graph visualizations
- Performed bugfixes and quality-of-life changes on widely used TA/NDA request web application built using **C#**, **ASP.NET** and a **SQL** database
- Eliminated time-consuming manual labor for Contracts department using **PowerShell** scripts to automatically update macro-enabled spreadsheets

Controls Subteam Member — *Blacksburg, VA*

VT BOLT - Electric Motorcycle Team — *Aug. 2019 - Present*

- Wrote **C** interface for TI C2000 DAC
- Used **C** to write performant memory-constrained (< 204KB RAM) 32-bit precision lookup tables for the TI C2000 microcontroller
- Created **Python** and **Matplotlib**-based GUI featuring 3-dimensional lookup table visualizations

Software Team Lead — *Haymarket, VA*

ILITE Robotics - FIRST Robotics Team 1885 — *Jun. 2017 - Jun. 2019*

- Designed and taught courses on **Java**programming, **Git**, cybersecurity
- Led development of **Java** software for feedback controls, state machines, logging mechanisms, and graphical interfaces for competition robots with team of 6-8 students
- 3-time District competition winner, District Championship winner, 2-time World Championship division quarterfinalist

Sales Associate — *Haymarket, VA*

Staples — *Jun. 2018 - Dec. 2019*

- Responsible for restocking shelves, cashiering, and helping customers

Volunteer Coach
Haymarket, VA

ILITE Robotics - FIRST Robotics Team 1885 — *Aug. 2015 - Jun. 2019*

- Taught STEM summer camps of 20 elementary and middle school students
- Created programming curriculum for Scratch and PBASIC
- Assisted with teaching CyberPatriot camp (Introduction to CyberSecurity)

Team Member - Windows Operating Systems
Haymarket, VA

AFA CyberPatriot Team — *Aug. 2015 - Jun. 2019*

- Worked with a team of 6 fellow high school students to perform OS hardening on Windows and Ubuntu OSes
- Wrote scripts to perform OS hardening on Windows operating systems using **Batch**, **PowerShell** to perform Registry and Group Policy configuration
- 2-time National Semifinalist – Platinum division

Projects

Multi-Agent Heron Manager

Heron Systems — *2021*

- **Python** framework enabling multi-agent interaction with multi-threaded and distributed compute capability
- Designed in collaboration with senior engineers and developed with 2 fellow interns
- Integrates with open-source and in-house reinforcement learning libraries
- Implements multi-agent air-to-air combat environment using JSBSim and RLLib
- Actively maintained in-house

FRC Robot Codebase

ILITE Robotics - FIRST Robotics Team 1885 — *2017-2019*

- Led development and Git version-control of 12,000+ line **Java** codebase integrated with **Gradle** and **Travis CI**
- Created **JavaFX** client-side GUIs for displaying robot odometry data in real-time, specifying constraints for autonomous decision-making, and editing system control parameters in real-time
- Used open-source libraries to implement robot trajectory generation and following using arc-approximated splines and high-frequency PID controllers in **Java**

Robot Arm

Personal Project — *2019*

- Designed and 3D printed low-cost 2 degree-of-freedom robot arm using SOLIDWORKS
- Wrote **C++** Arduino code receive commands from PC over serial and drive stepper motors
- Created **Qt** graphical interface in **C++** to control arm over serial
- Implemented inverse kinematics algorithm for arm

2D Space Sim

Personal Project — *2018-2019*

- Wrote a 2D space simulation with Newtonian physics in **Java** using LibGDX and Box2D libraries
- Created serializable "immediate mode" rendering configuration system using JSON file format

Driver Signaling System – FIRST Robotics Team 1885

ILITE Robotics - FIRST Robotics Team 1885 — *2016-2017*

- Created driver signaling system to display robot game and error states
- Used **Java** and **C++** to facilitate communication over an I2C bus between Linux robot controller, Arduino, and WS2812 addressable LEDs

Accomplishments

2018-2019	**Advanced Computer Studies Student of the Year**, Battlefield High School	*Haymarket, VA*
2018-2019	**Dean's List**, Northern Virginia Community College	*Haymarket, VA*
2017-2019	**National Semi-Finalist**, CyberPatriot	*Haymarket, VA*
Apr. 2018	**Presenter for *Effective Student Leadership in the FIRST Robotics Competition***, FRC World Championship	*Detroit, MI*
Apr. 2022	**Author for *A Mapping Approach to Achieve Torque Control for Parallel-Actuated Robotic Systems***, ASME 2022	*TBD*